Lab 2

By: Anderson Friday

Part I

Verilog Code

```
module main(
    input [9:0] SW,
    output [9:0] LEDR,

    input    MAX10_CLK1_50,
    output   [7:0]    HEX0,
    output   [7:0]    HEX1,
    output   [7:0]    HEX2,
    output   [7:0]    HEX3,
    output   [7:0]    HEX4,
    output   [7:0]    HEX5
);
    // Part I
    assign LEDR = SW;

     // Part I: Decode 4-bit switch inputs to 7-segment displays
    Seg7_Decoder dec0 (.m(SW[3:0]), .out(HEX0));
    Seg7_Decoder dec1 (.m(SW[7:4]), .out(HEX1));

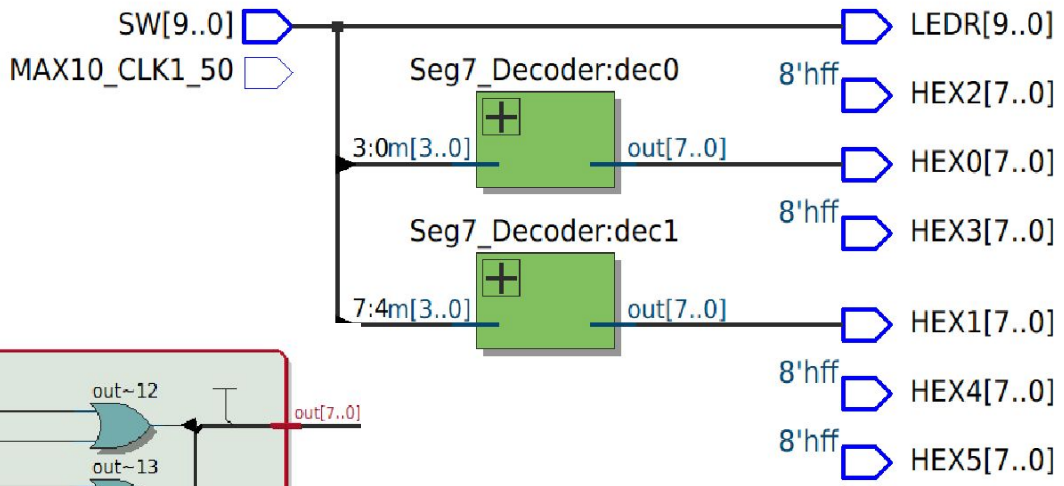
     // Turn off unused displays (active-low segments: 8'hFF = all OFF)
    assign HEX2 = 8'hFF;
    assign HEX3 = 8'hFF;
    assign HEX4 = 8'hFF;
    assign HEX5 = 8'hFF;
```

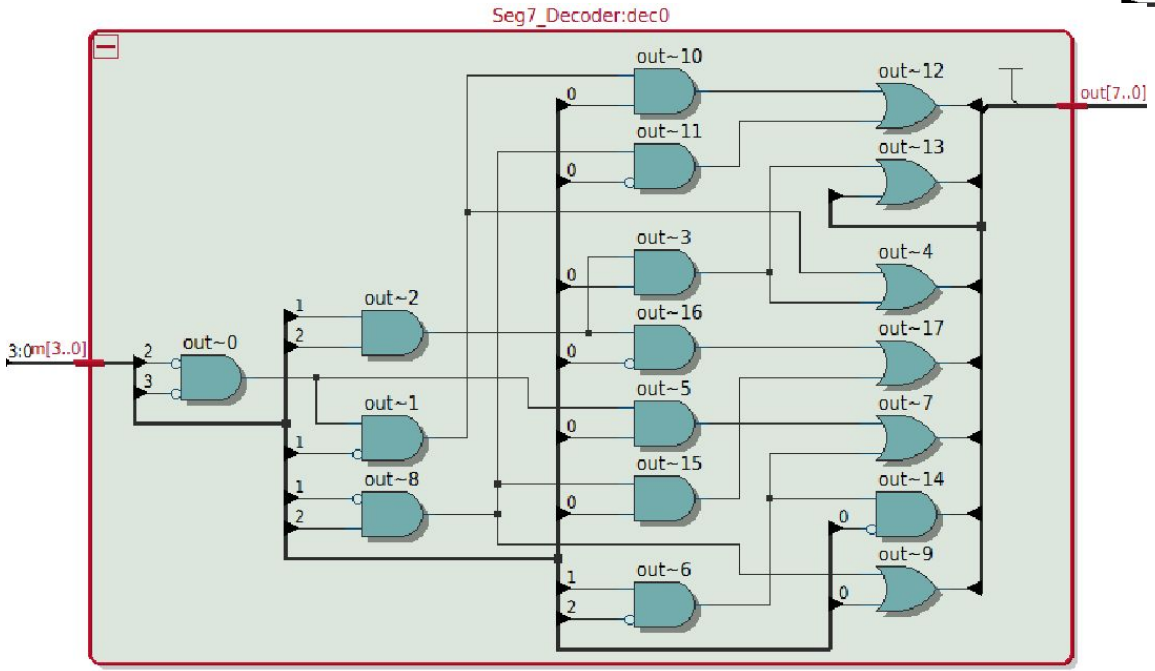
```
module Seg7_Decoder(
    input [3:0] m,
    output [7:0] out
);

    assign out[7] = 1'b1;
    assign out[6] = ~m[3]&~m[2]&~m[1]|m[2]&m[1]&m[0];
    assign out[5] = ~m[3]&~m[2]&m[0]|~m[2]&m[1];
    assign out[4] = m[0]|m[2]&~m[1];
    assign out[3] = ~m[3]&~m[2]&~m[1]&m[0]|m[2]&~m[1]&~m[0]|m[2]&m[1]&m[0];
    assign out[2] = ~m[2]&m[1]&~m[0];
    assign out[1] = m[2]&~m[1]&m[0]|m[2]&m[1]&~m[0];
    assign out[0] = ~m[3]&~m[2]&~m[1]&m[0]|m[2]&~m[1]&~m[0];

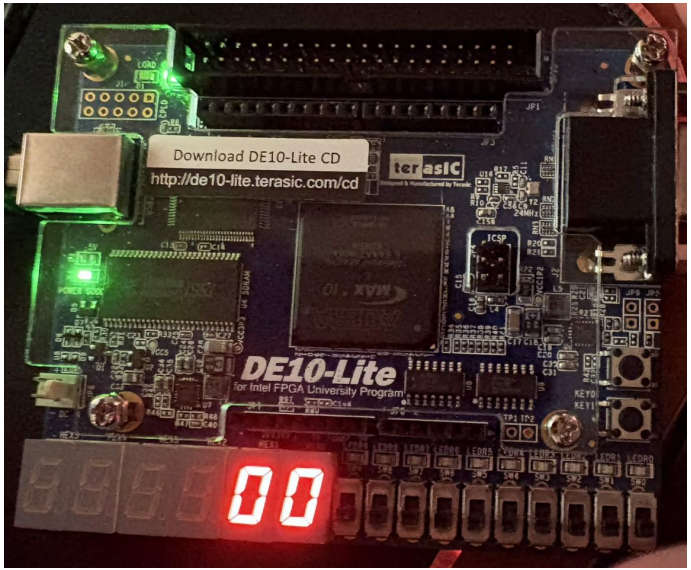
endmodule
```

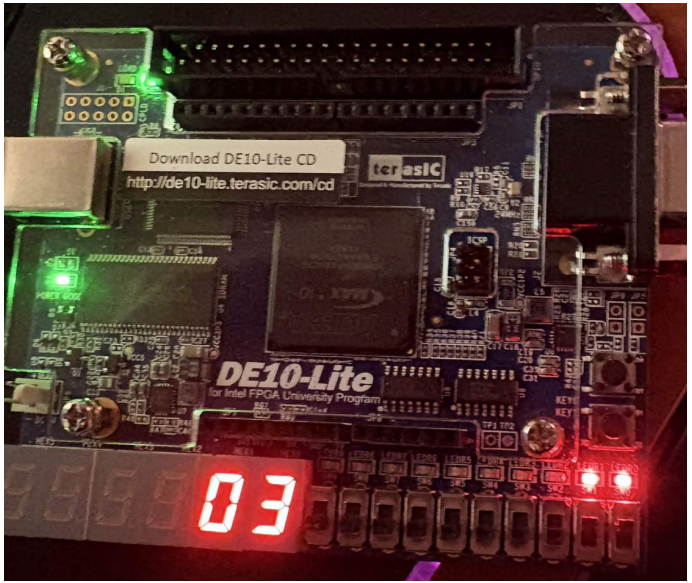
RTL Viewer


SW[9..0]
MAX10_CLK1_50
LEDR[9..0]
Seg7_Decoder:dec0
3:0m[3..0]
out[7..0]
8'hff
HEX2[7..0]
HEX0[7..0]
8'hff
HEX3[7..0]
Seg7_Decoder:dec1
7:4m[3..0]
out[7..0]
HEX1[7..0]
8'hff
HEX4[7..0]
8'hff
HEX5[7..0]


Seg7_Decoder:dec0
3:0m[3..0]
out~0
out~1
out~2
out~8
out~3
out~4
out~5
out~6
out~7
out~9
out~10
out~11
out~12
out~13
out~14
out~15
out~16
out~17
out[7..0]

DE10-Lite Board





Part 2

Verilog Code

```
// PART II
wire [3:0] w_m;

// 7 Segment Display
Seg7_Decoder D0(.m(w_m), .out(HEX0));

wire w_z;
wire [2:0] w_ca;

// MUltiplexers
mux_2_1 M0(.s(w_z), .x(SW[0]), .y(w_ca[0]), .m(w_m[0]));
mux_2_1 M1(.s(w_z), .x(SW[1]), .y(w_ca[1]), .m(w_m[1]));
mux_2_1 M2(.s(w_z), .x(SW[2]), .y(w_ca[2]), .m(w_m[2]));
mux_2_1 M3(.s(w_z), .x(SW[3]), .y(1'b0), .m(w_m[3]));

// Circuit A
CircuitA CA(.v({SW[2], SW[1], SW[0]}), .out(w_ca));

// Comparator
Comparator myC(.v({SW[3], SW[2], SW[1], SW[0]}), .z(w_z));

// Circuit B
CircuitB CB( .z(w_z), .s(HEX1));
```

```
module CircuitB(
    input z,
    output [7:0] s
);
    assign s[7] = 1'b1;
    assign s[6] = 1'b1;
    assign s[5] = z;
    assign s[4] = z;
    assign s[3] = z;
    assign s[2] = 1'b0;
    assign s[1] = 1'b0;
    assign s[0] = z;

endmodule
```

```
module CircuitA(
    input [2:0] v,
    output [2:0] out
);

    assign out[2] = v[2]&v[1];
    assign out[1] = ~v[1];
    assign out[0] = v[0];

endmodule
```

```
module mux_2_1(
    input s,
    input x,
    input y,
    output m
);

    assign m = (~s&x)|(s&y);

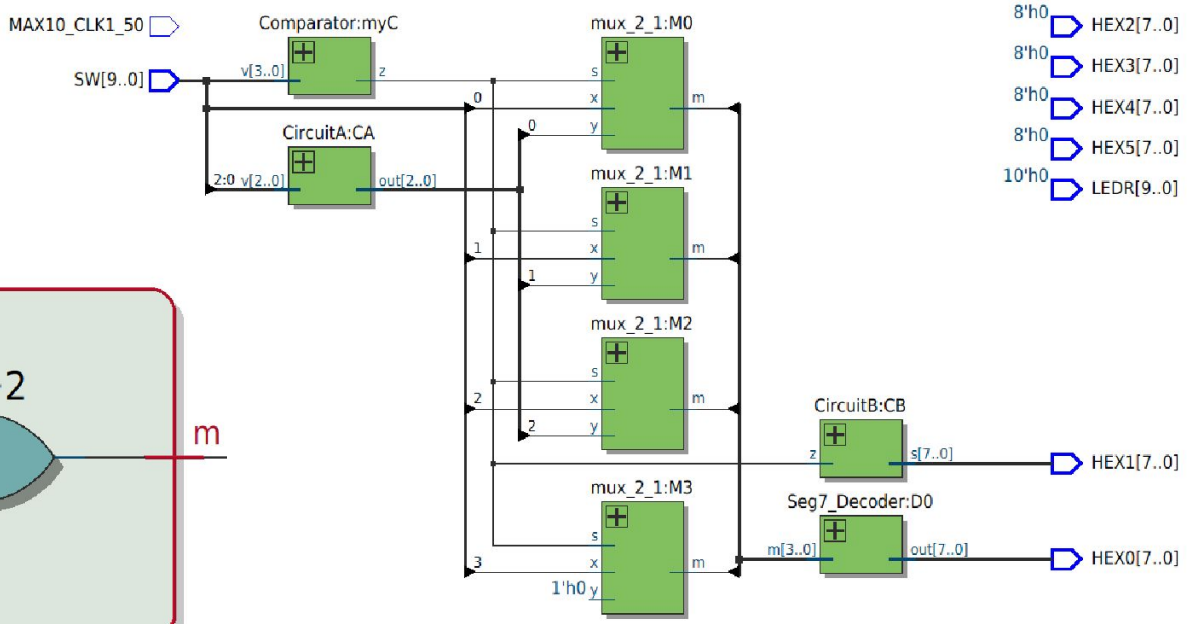
endmodule
```

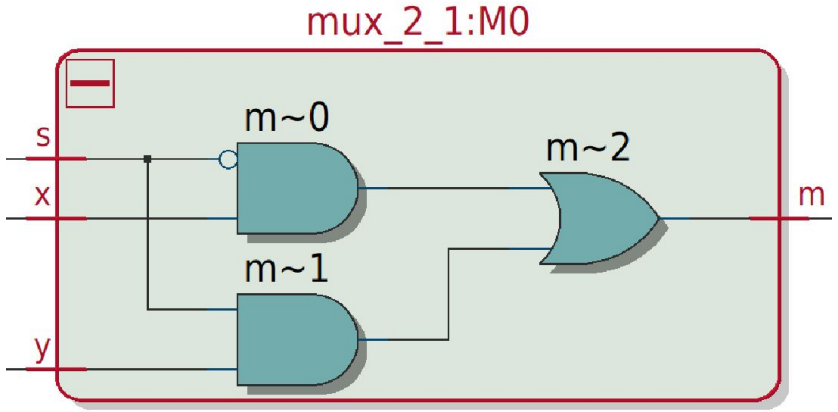
```
module Comparator(
    input [3:0] v,
    output z
);

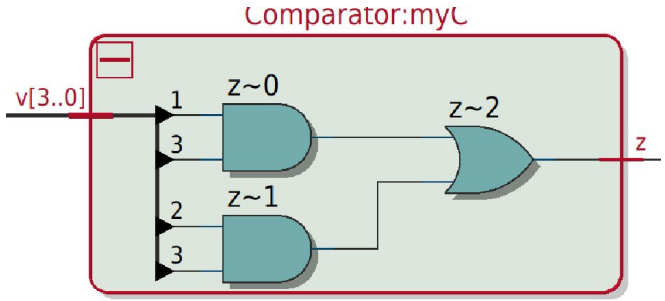
    assign z = v[3]&v[1]|v[3]&v[2];

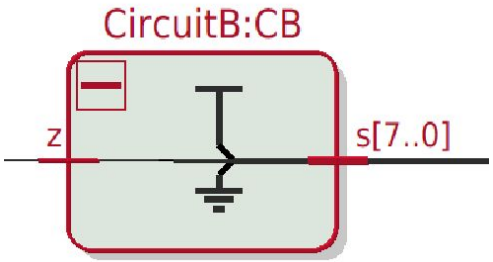
endmodule
```

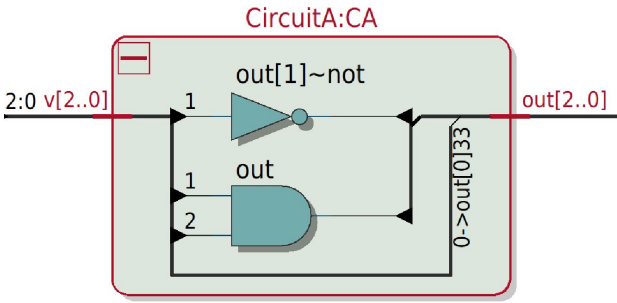
RTL Viewer


MAX10_CLK1_50
SW[9..0]
Comparator:myC
v[3..0]
z
CircuitA:CA
2:0 v[2..0]
out[2..0]
mux_2_1:M0
mux_2_1:M1
mux_2_1:M2
mux_2_1:M3
s
x
y
m
1'h0
CircuitB:CB
s[7..0]
Seg7_Decoder:D0
m[3..0]
out[7..0]
8'h0
HEX2[7..0]
HEX3[7..0]
HEX4[7..0]
HEX5[7..0]
10'h0
LEDR[9..0]
HEX1[7..0]
HEX0[7..0]

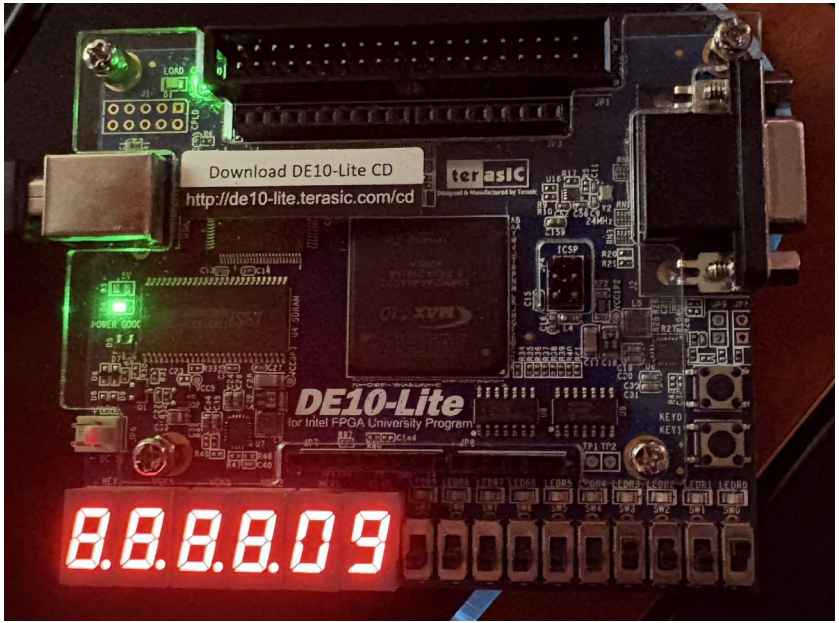

mux_2_1:M0
s
x
y
m~0
m~1
m~2
m

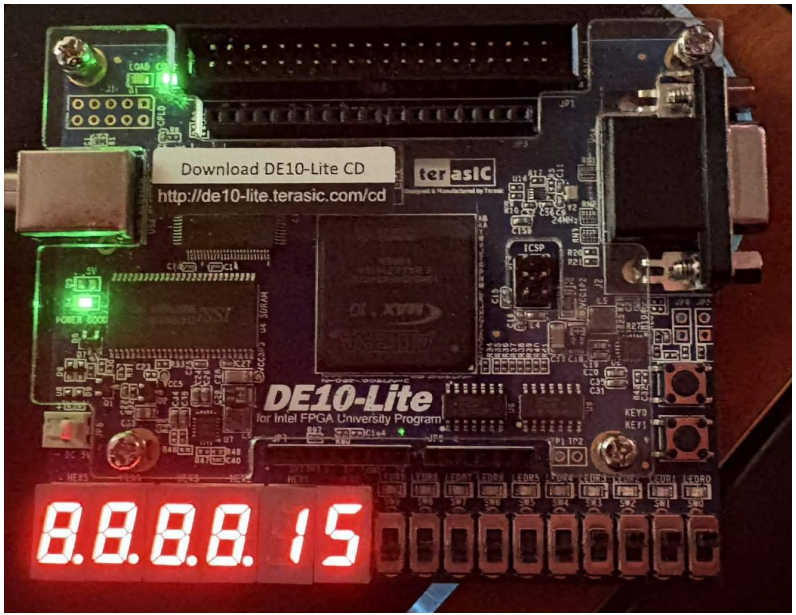

Comparator:myC
v[3..0]
z~0
z~1
z~2
z


CircuitB:CB
z
s[7..0]


CircuitA:CA
2:0 v[2..0]
out[1]~not
out
out[2..0]
0->out[0]33

DE10-Lite Board





Part 3

Verilog Code

```
//Part III
wire [3:0] s;
wire c_o;

assign LEDR[8:0] = SW[8:0];
assign LEDR[9] = c_o;

rip_carry_adder_4bit rca (.a(SW[7:4]), .b(SW[3:0]), .c_i(SW[8]), .s(s), .c_o(c_o));
Seg7_Decoder dec0 (.m(s), .out(HEX0));
```

```
module full_adder(
    input a, b, c_i,
    output s, c_o
);

    wire ab_xor;

    assign ab_xor = a ^ b;
    assign s = ab_xor ^ c_i;

    mux_2_1 mux (.s(ab_xor), .x(b), .y(c_i), .m(c_o));

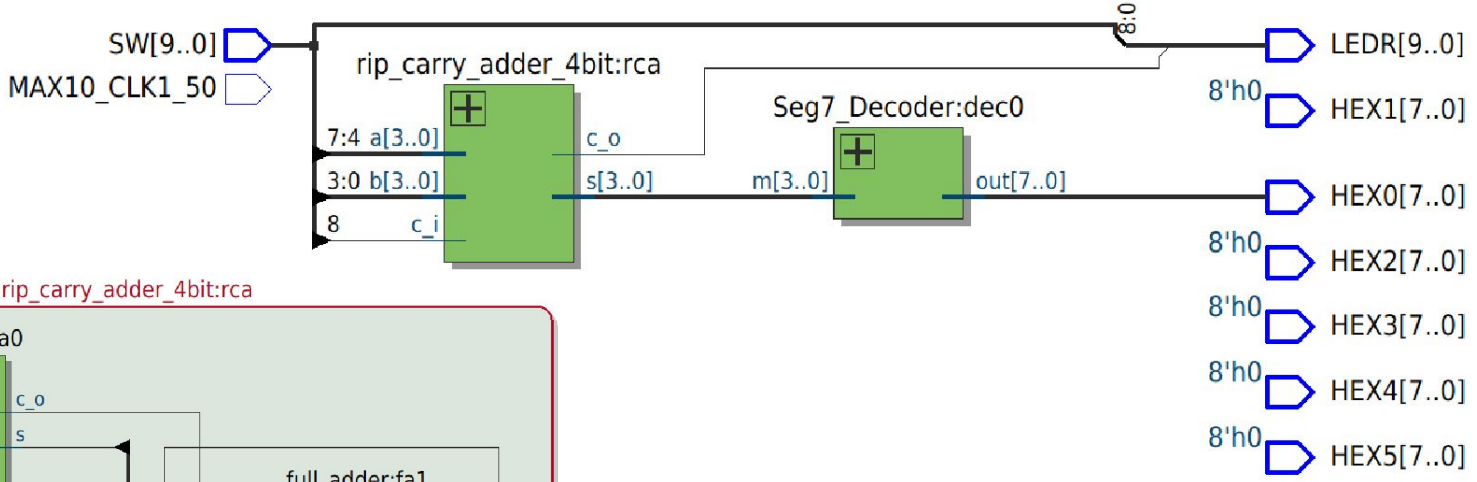
endmodule
```

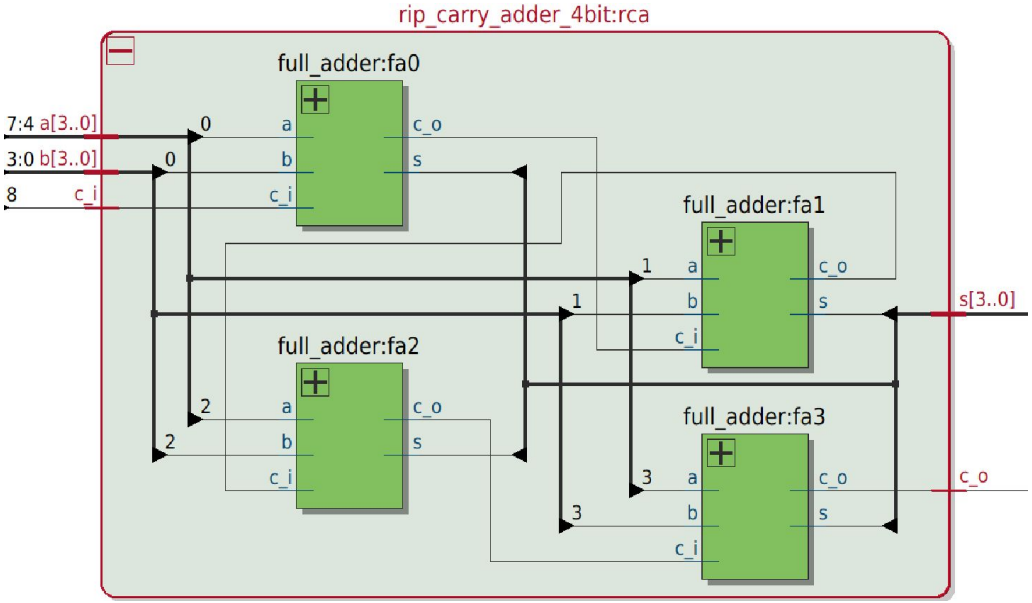
```
module rip_carry_adder_4bit (
    input [3:0] a, b,
    input c_i,
    output [3:0] s,
    output c_o
);

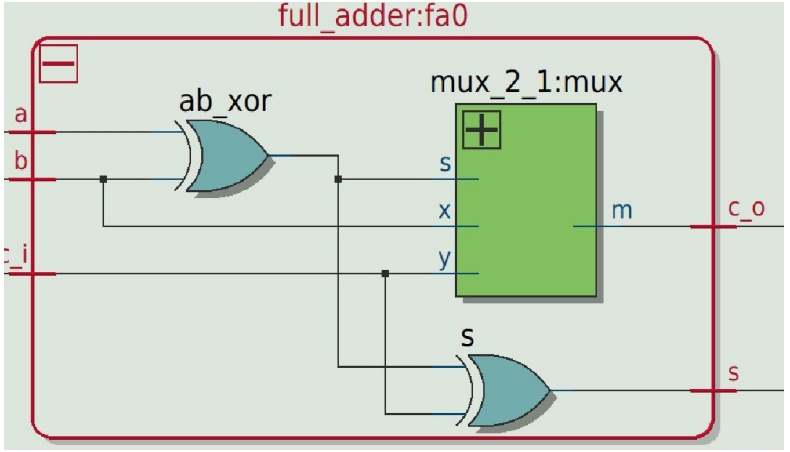
    wire c1, c2, c3;
    full_adder fa0 (.a(a[0]), .b(b[0]), .c_i(c_i), .s(s[0]), .c_o(c1));
    full_adder fa1 (.a(a[1]), .b(b[1]), .c_i(c1), .s(s[1]), .c_o(c2));
    full_adder fa2 (.a(a[2]), .b(b[2]), .c_i(c2), .s(s[2]), .c_o(c3));
    full_adder fa3 (.a(a[3]), .b(b[3]), .c_i(c3), .s(s[3]), .c_o(c_o));

endmodule
```

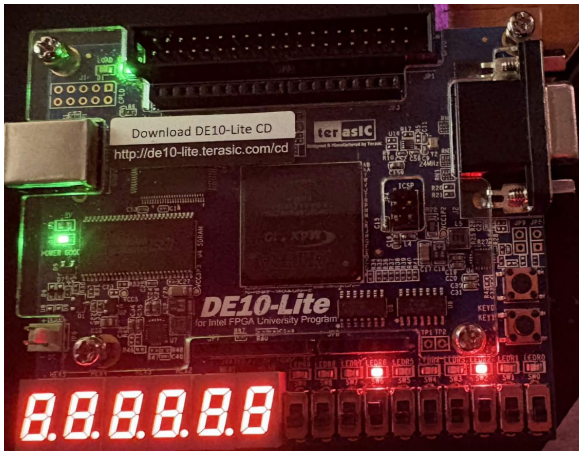
RTL Viewer

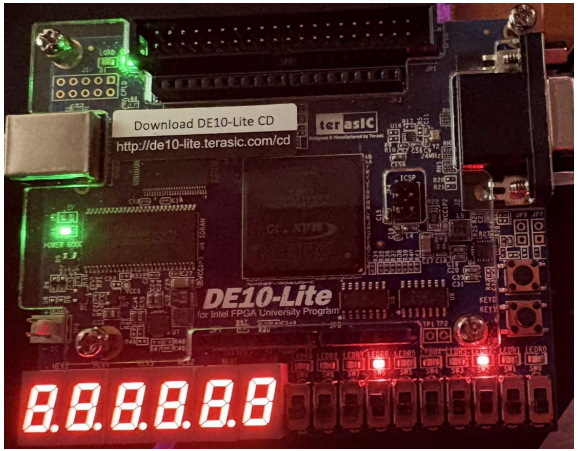

SW[9..0]
MAX10_CLK1_50
rip_carry_adder_4bit:rca
7:4 a[3..0]
3:0 b[3..0]
8 c_i
c_o
s[3..0]
8:0
Seg7_Decoder:dec0
m[3..0]
out[7..0]
LEDR[9..0]
8'h0
HEX1[7..0]
HEX0[7..0]
HEX2[7..0]
HEX3[7..0]
HEX4[7..0]
HEX5[7..0]

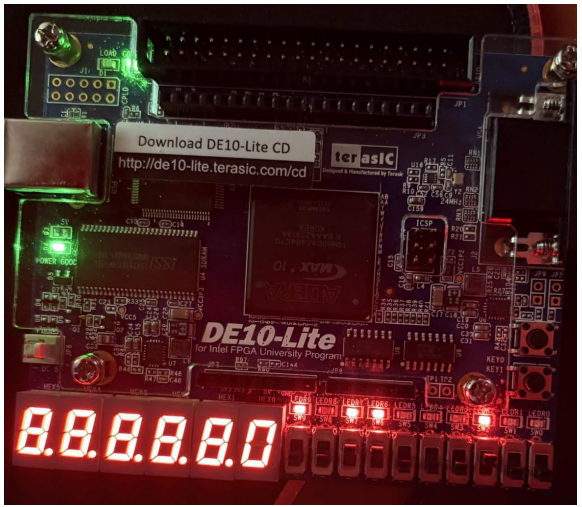

rip_carry_adder_4bit:rca
full_adder:fa0
full_adder:fa1
full_adder:fa2
full_adder:fa3
7:4 a[3..0]
3:0 b[3..0]
8 c_i
a
b
c_i
c_o
s
s[3..0]


full_adder:fa0
ab_xor
mux_2_1:mux
a
b
c_i
s
x
y
m
c_o
s

DE10-Lite Board







Part 4

Verilog Code

```
//Part IV

    wire [3:0] A, B, sum, S0;
    wire cin, carry, z;

    assign A   = SW[7:4];
    assign B   = SW[3:0];
    assign cin = SW[8];

    assign LEDR[8:0] = SW[8:0];
    assign LEDR[9]   = (A[3] & (A[2] | A[1])) | (B[3] & (B[2] | B[1]));

    wire [3:0] a_out;
    CircuitA CA (.carry(carry), .sum(sum), .a_out(a_out));
    Comparator comp (.carry(carry), .sum(sum), .z(z));
    rip_carry_adder_4bit adder (.a(A), .b(B), .c_i(cin), .s(sum), .c_o(carr

    assign S0[0] = (~z & sum[0]) | (z & a_out[0]);
    assign S0[1] = (~z & sum[1]) | (z & a_out[1]);
    assign S0[2] = (~z & sum[2]) | (z & a_out[2]);
    assign S0[3] = (~z & sum[3]) | (z & a_out[3]);

    assign S1 = {3'b000, z};

    Seg7_Decoder dec_S0 (.m(S0), .out(HEX0));  // Ones digit of sum
    Seg7_Decoder dec_S1 (.m(S1), .out(HEX1));
    Seg7_Decoder dec_B  (.m(B),  .out(HEX3));  // Input B
    Seg7_Decoder dec_A  (.m(A),  .out(HEX5));  // Input A

    assign HEX2 = 8'hFF;
    assign HEX4 = 8'hFF;
```

```
//Part IV
module Comparator(
    input [3:0] sum,
    input carry,
    output z
);

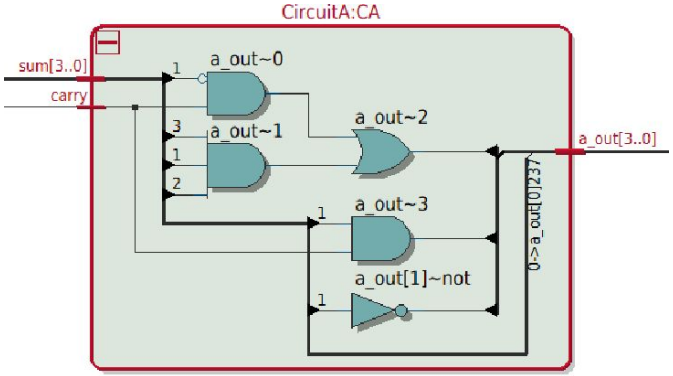
    assign z = carry | (sum[3] & sum[2]) | (sum[3] & sum[1]);

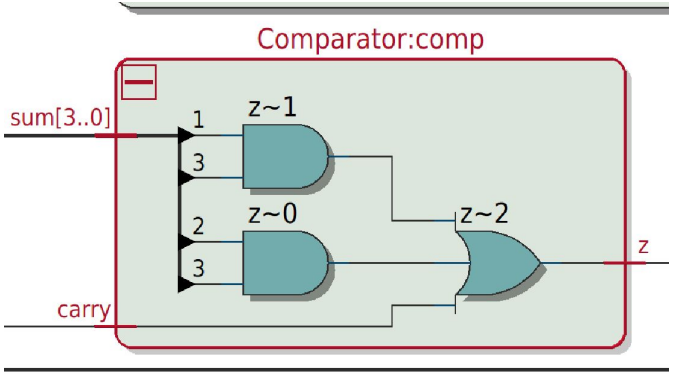
endmodule
```

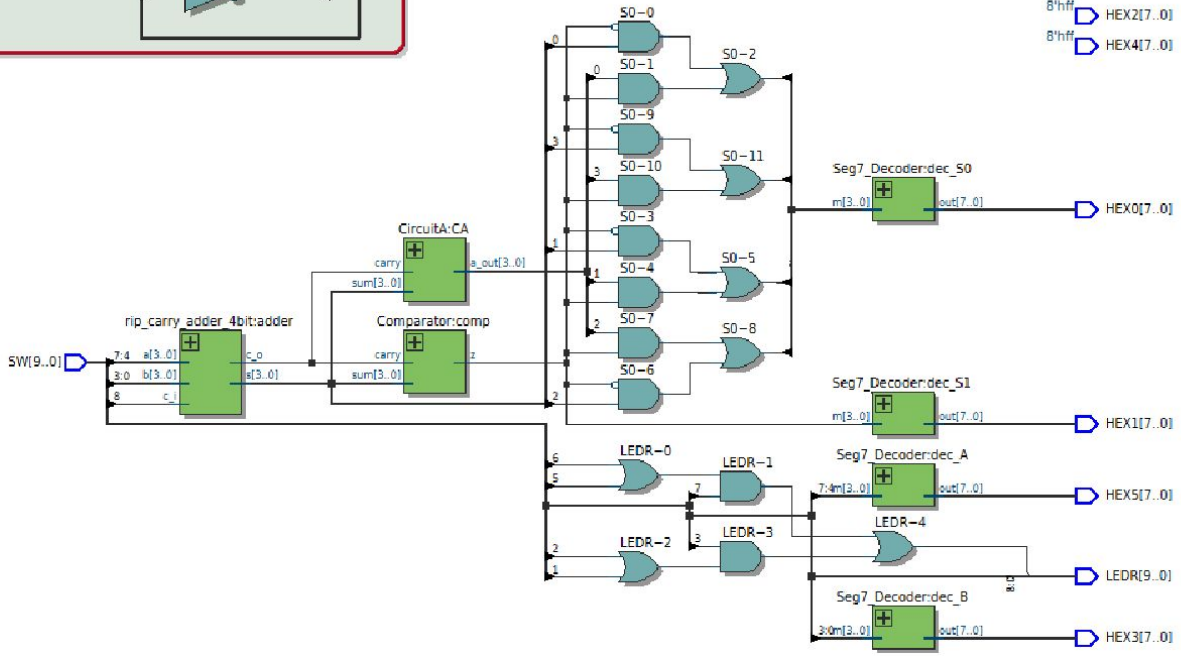
```
//Part IV
module CircuitA(
    input [3:0] sum,
    input carry,
    output [3:0] a_out
);

    assign a_out[0] = sum[0];
    assign a_out[1] = ~sum[1];
    assign a_out[2] = (carry & ~sum[1]) | (sum[1] & sum[2] & sum[3]);
    assign a_out[3] = carry & sum[1];
endmodule
```

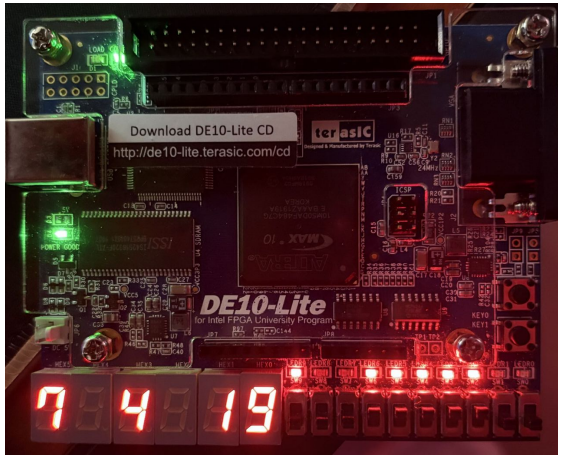
RTL Viewer

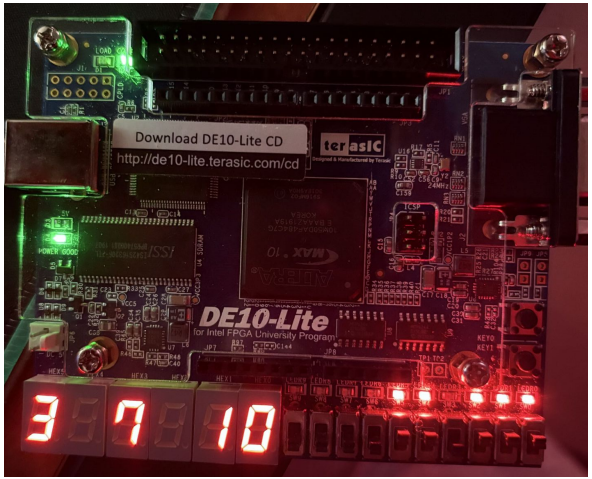

CircuitA:CA
sum[3..0]
carry
a_out~0
a_out~1
a_out~2
a_out~3
a_out[1]~not
a_out[3..0]
0->a_out[0]237

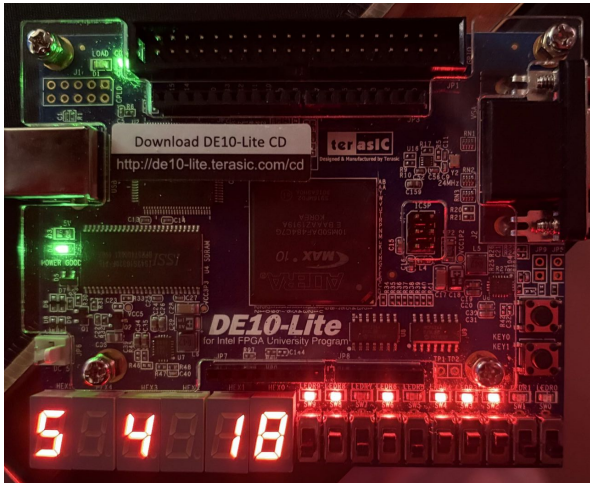

Comparator:comp
sum[3..0]
carry
z~1
z~0
z~2
z


SW[9..0]
rip_carry_adder_4bit:adder
CircuitA:CA
Comparator:comp
S0~0
S0~1
S0~2
S0~3
S0~4
S0~5
S0~6
S0~7
S0~8
S0~9
S0~10
S0~11
Seg7_Decoder:dec_S0
Seg7_Decoder:dec_S1
Seg7_Decoder:dec_A
Seg7_Decoder:dec_B
LEDR~0
LEDR~1
LEDR~2
LEDR~3
LEDR~4
8'hff
HEX2[7..0]
8'hff
HEX4[7..0]
HEX0[7..0]
HEX1[7..0]
HEX5[7..0]
LEDR[9..0]
HEX3[7..0]

DE10-Lite Board







Part 5

Verilog Code

```
//Part V

   wire [3:0] A0, B0, A1, B1, S0, S1, S2;
   wire c1, c2;

   assign A0 = SW[7:4];
   assign B0 = SW[3:0];
   assign A1 = {3'b000, SW[8]};
   assign B1 = {3'b000, SW[9]};

   assign LEDR = SW;
   assign S2 = {3'b000, c2};

   bcd_adder bcd0 (.A(A0), .B(B0), .c_i(1'b0), .s(S0), .c_o(c1));
   bcd_adder bcd1 (.A(A1), .B(B1), .c_i(c1), .s(S1), .c_o(c2));

   Seg7_Decoder disp_S0 (.m(S0), .out(HEX0));
   Seg7_Decoder disp_S1 (.m(S1), .out(HEX1));
   Seg7_Decoder disp_S2 (.m(S2), .out(HEX2));

endmodule
```

```
module bcd_adder (
    input  wire [3:0] A,
    input  wire [3:0] B,
    input  wire       c_i,
    output wire [3:0] s,
    output wire       c_o
);

    wire [3:0] sum;
    wire [3:0] a_out;
    wire       z;
    wire       carry;

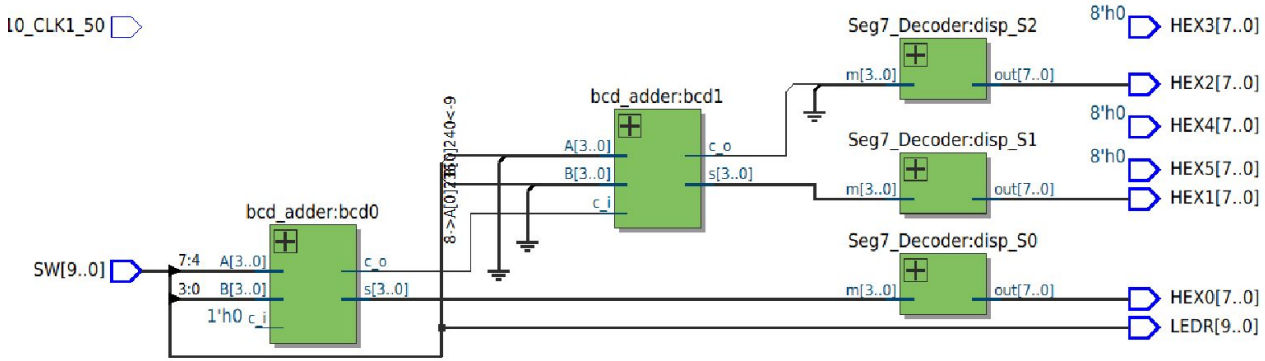
    rip_carry_adder_4bit (.a(A), .b(B), .c_i(c_i), .s(sum), .c_o(carry));
    Comparator compare_bcd (.carry(carry), .sum(sum), .z(z));
    CircuitA convert_bcd (.carry (carry), .sum   (sum), .a_out (a_out));

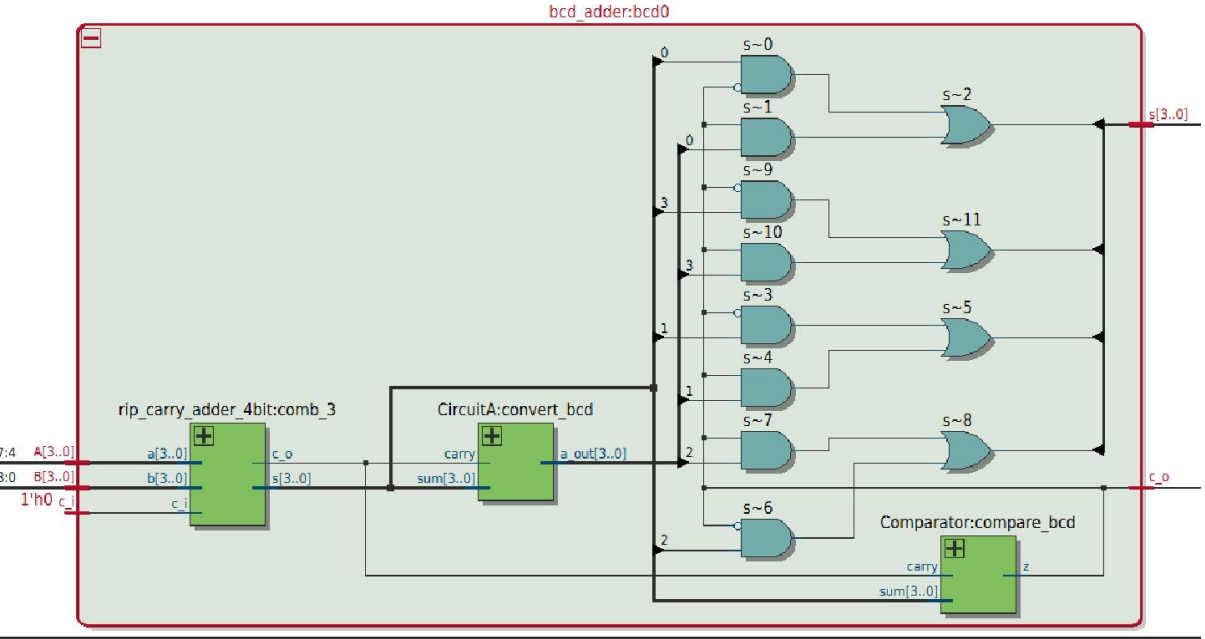
    assign s[0] = (~z & sum[0]) | (z & a_out[0]);
    assign s[1] = (~z & sum[1]) | (z & a_out[1]);
    assign s[2] = (~z & sum[2]) | (z & a_out[2]);
    assign s[3] = (~z & sum[3]) | (z & a_out[3]);

    assign c_o = z;

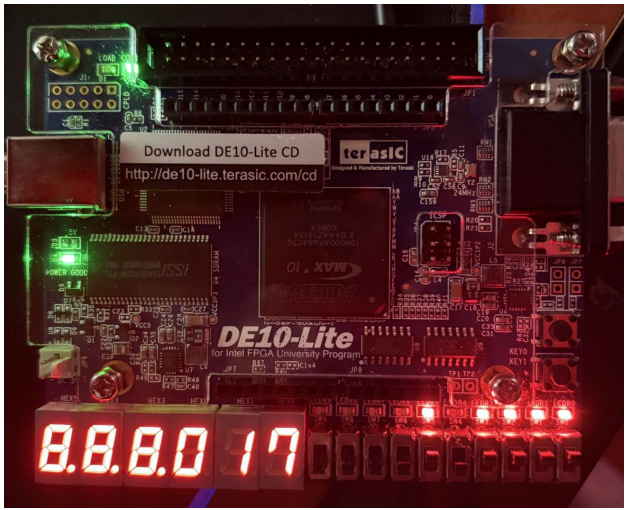
endmodule
```

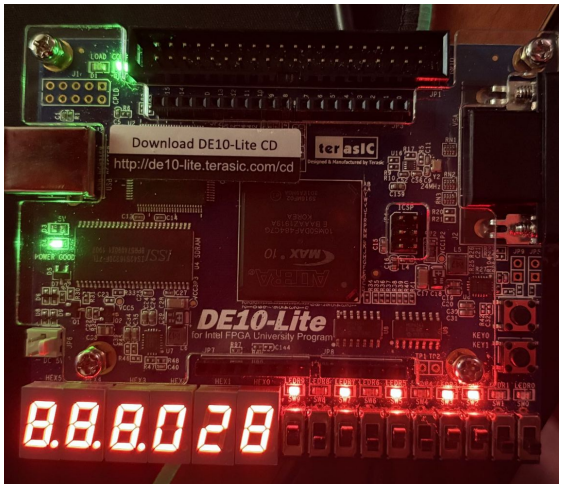
RTL Viewer

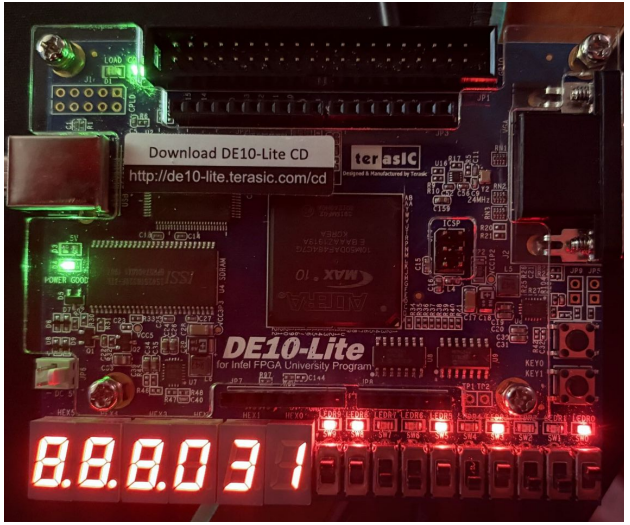

l0_CLK1_50
SW[9..0]
bcd_adder:bcd0
bcd_adder:bcd1
Seg7_Decoder:disp_S2
Seg7_Decoder:disp_S1
Seg7_Decoder:disp_S0
8'h0
HEX3[7..0]
HEX2[7..0]
HEX4[7..0]
HEX5[7..0]
HEX1[7..0]
HEX0[7..0]
LEDR[9..0]


bcd_adder:bcd0
rip_carry_adder_4bit:comb_3
CircuitA:convert_bcd
Comparator:compare_bcd
s~0
s~1
s~2
s~3
s~4
s~5
s~6
s~7
s~8
s~9
s~10
s~11
s[3..0]
c_o

DE10-Lite Board







Made alterations since DE10 doesn't have 15 SW. SW 8 and 9 add 10 to respected input as an alternative way to display a higher output. However had issues displaying it correctly as there was some logical errors in the bcd